

Smart Home Temperature Agent

AI Agent Design & PEAS Evaluation — Lab Assignment

1. Agent Overview

Agent Name: ThermoSense — a Smart Home Temperature Agent.

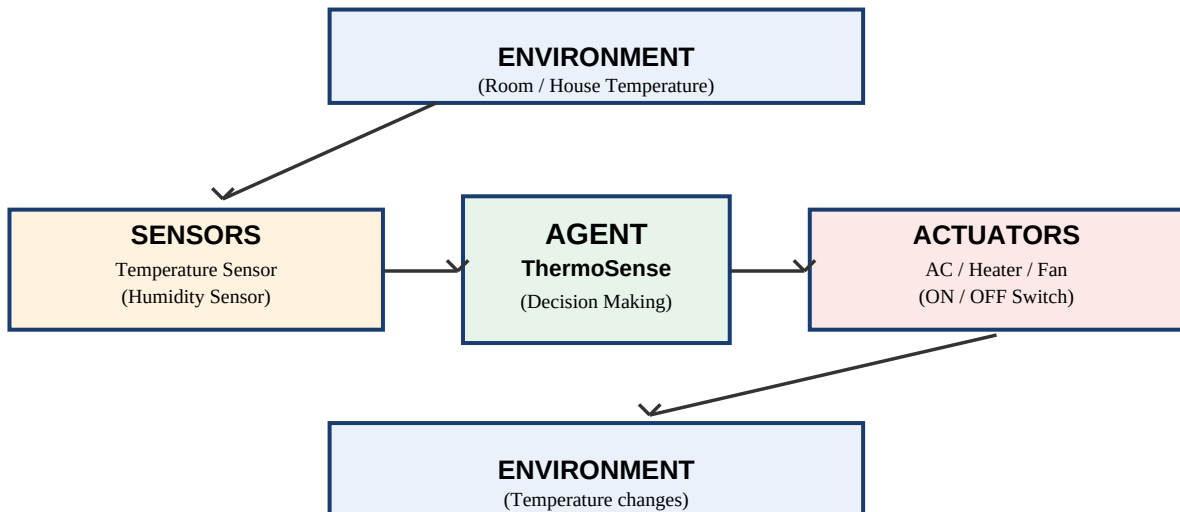
ThermoSense continuously monitors the temperature of a room/house and automatically controls the AC, heater, and fan to keep the room at a comfortable temperature, without needing a human to manually operate the appliances.

Simple Rule Set used by the Agent:

Condition (Sensor Reading)	Action (Actuator)
Temperature = 30 C (Hot)	AC -> ON
Temperature = 24 C (Comfortable)	AC -> OFF
Temperature = 18 C (Cold)	Heater -> ON

2. Graphical Representation

The diagram below shows how the agent continuously interacts with its environment in a perceive → decide → act cycle.



3. Agent Formulation

3.1 Environment (According to Types)

Property	Type	Justification
Observability	Partially Observable	The sensor gives the current temperature reading, but the agent cannot sense everything (e.g. whether a window/door is open, or how many people are in the room).
Determinism	Stochastic	Temperature is affected by unpredictable outside factors like weather, sunlight, or number of people, so the same action does not always give the same result.
Episodic / Sequential	Sequential	The current action (e.g. turning the AC ON) affects future temperature readings and future decisions.
Static / Dynamic	Dynamic	The room temperature keeps changing even while the agent is deciding what to do.
Discrete / Continuous	Continuous	Temperature values are continuous (e.g. 24.5 degree C), not fixed, limited steps.
Agents	Single-Agent	Only this one agent is responsible for controlling the house temperature.

3.2 Sensors

- Temperature sensor – continuously measures the current room temperature.
- Humidity sensor (optional) – measures moisture level in the air for extra comfort control.

3.3 Actuators

- Air Conditioner – switched ON/OFF to cool the room.
- Heater – switched ON/OFF to warm the room.
- Fan – switched ON/OFF for air circulation.

3.4 Agent Type

Simple Reflex Agent. The agent selects its action using direct condition-action (if-then) rules based only on the current temperature reading — it does not need to remember past readings to decide the next action (e.g. "IF temperature = 30 C THEN turn AC ON"). This matches the classic Simple Reflex Agent pattern, similar to a thermostat.

4. PEAS Framework Evaluation

Component	Description
Performance Measure	Maintains a comfortable room temperature, minimizes unnecessary energy usage, and responds quickly and accurately to temperature changes.

Environment	The room / house interior — including current temperature, humidity, and the people present in it.
Actuators	Air Conditioner (ON/OFF), Heater (ON/OFF), Fan (ON/OFF).
Sensors	Temperature sensor (and optionally a humidity sensor) that continuously reads the current room conditions.

5. Rationality Explanation

An agent is considered **rational** if, for every possible perception sequence, it selects an action that is expected to maximize its performance measure, given the information provided by its sensors and whatever built-in knowledge it has.

ThermoSense is rational because: whenever its temperature sensor perceives a value that moves away from the comfortable range, it immediately chooses the action (AC ON, Heater ON, or turning appliances OFF) that is expected to bring the room back to a comfortable temperature while avoiding unnecessary energy usage — which is exactly what its Performance Measure (comfort + energy saving) requires. It does not take random or wasteful actions (e.g. it does not run the AC when the temperature is already comfortable), so its behaviour is consistently aligned with its goal.